MBSD Exam Preparation Guide

Excellent 👍
Handshaking in 8255 is a **teacher-favorite, diagram-based, high-scoring topic**.
If you explain **signals + timing + role of Port C**, you almost *guarantee full marks*.

Below is a **COMPLETE EXAM-READY ANSWER** in **pure book language**, exactly the way your teacher expects.

HANDSHAKING IN 8255 PROGRAMMABLE PERIPHERAL INTERFACE

Introduction (WRITE THIS – VERY IMPORTANT)

In a microprocessor-based system, peripheral devices generally operate at **slower speeds** compared to the CPU.
To ensure **reliable and synchronized data transfer**, a mechanism is required to coordinate communication between the CPU and I/O devices.

This mechanism is known as **handshaking**.

In the **8255 Programmable Peripheral Interface**, handshaking is implemented using **Mode 1 and Mode 2**, where **Port C lines** are used as **control and status signals**.

Definition (BOOK LANGUAGE)

Handshaking is a method of data transfer in which **control signals are exchanged between the CPU and the peripheral device** to indicate readiness, data availability, and completion of data transfer.

Why Handshaking is Required (EXAM KEYWORDS)

- Peripheral devices are **slow**
- Prevents **data loss**
- Ensures **synchronization**

- Provides **error-free data transfer**
- Enables **interrupt-driven operation**

Handshaking in 8255

Handshaking is used in:

- **Mode 1 – Strobed Input / Output**
- **Mode 2 – Bi-directional Bus**

👉 Handshaking signals are provided through **Port C**.

MODE 1 HANDSHAKING (MOST IMPORTANT)

Mode 1 supports:

- **Strobed Input**
- **Strobed Output**

A) MODE 1 – STROBED INPUT HANDSHAKING

Block Diagram (DRAW THIS IN EXAM)

```lua
Peripheral Device
    |
    | STB‾
    v
+---------------+
|    8255       |
|               |
| Port A / B    | ---> CPU
|               |
| Port C        |
| (Handshake)   |
+---------------+
```

Handshaking Signals (WRITE AS TABLE)

Signal	Port C Line	Function
$STB^{\overline{\ }}$	PC4 / PC2	Strobe input from device
IBF	PC5 / PC1	Input Buffer Full
INTR	PC6 / PC3	Interrupt Request

Working (STEP-BY-STEP – VERY IMPORTANT)

1. Peripheral device places data on the data bus.
2. Device activates **$STB^{\overline{\ }}$ (Strobe)** to indicate data is valid.
3. 8255 latches the data into the input buffer.
4. **IBF (Input Buffer Full)** becomes high.
5. If interrupts are enabled, **INTR** is generated.
6. CPU reads the data.
7. IBF and INTR are cleared automatically.

Timing Diagram (EXPLAIN IN WORDS – DIAGRAM OPTIONAL)

- Data must be stable before $STB^{\overline{\ }}$ goes low.
- IBF goes high after data is latched.
- INTR is generated if enabled.
- Data is cleared after CPU read.

Key Points (EXAM GOLD)

✔️ Data is latched
✔️ Reliable input operation
✔️ Interrupt supported

B) MODE 1 – STROBED OUTPUT HANDSHAKING

Block Diagram

```sql
    CPU
     |
     | OUT
     v
+---------------+
|    8255       |
|               |
| Port A / B    | ---> Peripheral
|               |
| Port C        |
| (Handshake)   |
+---------------+
```

Handshaking Signals

Signal	Port C Line	Function
$\overline{OBF}$	PC7 / PC1	Output Buffer Full
$\overline{ACK}$	PC6 / PC0	Acknowledge from device
INTR	PC5 / PC2	Interrupt Request

Working (WRITE THIS EXACTLY)

1. CPU writes data to output port.
2. 8255 sets **$\overline{OBF}$ low**, indicating output data is available.
3. Peripheral device reads the data.
4. Peripheral sends **$\overline{ACK}$** to 8255.
5. $\overline{OBF}$ is cleared.
6. INTR is generated if enabled.

MODE 2 HANDSHAKING (BI-DIRECTIONAL)

Definition (VERY IMPORTANT)

Mode 2 provides **bi-directional data transfer** on **Port A**, using handshaking signals from **Port C**.

Handshaking Signals in Mode 2

Signal	Function
$\overline{STB}$	Input strobe
IBF	Input buffer full
$\overline{OBF}$	Output buffer full
$\overline{ACK}$	Acknowledge
INTR	Interrupt

Block Diagram

```lua
        CPU
         |
         |
    +----------------+
    |     8255       |
    |                |
    |   Port A       | <----> Peripheral
    |                |
    |   Port C       |
    | (Handshake)    |
    +----------------+
```

Features of Mode 2

- Full duplex communication
- High-speed transfer
- Interrupt driven
- Complex control

Role of Port C in Handshaking (VERY IMPORTANT)

Port C is:

- Divided into **upper and lower halves**
- Used as:
  - Status signals
  - Control signals
  - Interrupt signals

👉 **Without Port C, handshaking is not possible in 8255.**

Comparison: Mode 0 vs Mode 1

Feature	Mode 0	Mode 1
Handshaking	No	Yes
Data reliability	Low	High
Interrupt support	No	Yes
Applications	Simple I/O	Keyboard, printer

Advantages of Handshaking

- Prevents data loss
- Synchronizes devices
- Supports interrupts
- Suitable for slow peripherals

Applications (EXAM LINE)

Handshaking in 8255 is widely used in **printer interfacing, keyboard input systems, industrial control, and data acquisition systems**.

SHORT ANSWER (3–4 MARKS)

> Handshaking in 8255 uses Port C lines to coordinate data transfer between CPU and peripheral devices. It ensures reliable data transfer using control signals such as STB, IBF, OBF, ACK, and INTR.

EXAM WRITING TIPS (VERY IMPORTANT)

✔️ Always write **signal names**
✔️ Explain **role of Port C**
✔️ Mention **interrupt support**
✔️ Draw **one block diagram minimum**

Next 🔥

Reply with 5️⃣ **Programmable Interrupt Controller (8259)**
or
6️⃣ **Double Buffering Solution**

Both are **high-probability final exam questions**, and we'll handle them **book-perfect**.

Perfect 💯
This is the **most scoring + least risky** part of MBSD finals.
If you draw a **clean block diagram** and explain it in **book language**, the examiner *cannot* cut marks.

Below are **TWO COMPLETE EXAM-READY APPLICATION ANSWERS** exactly at **5th-semester level**, with **clear diagrams + explanation**.

APPLICATIONS OF MICROPROCESSOR-BASED SYSTEMS

1️⃣ AIR QUALITY INDEX (AQI) MONITORING SYSTEM

Introduction (WRITE THIS)

Air pollution has become a major environmental concern affecting human health and ecosystem balance.
To monitor pollution levels and ensure public safety, a **microprocessor-based Air Quality Index (AQI) Monitoring System** is used.

This system measures various atmospheric parameters and computes the **Air Quality Index** to indicate pollution levels.

Block Diagram of AQI System (MUST DRAW)

```sql
+---------------+
| Gas Sensors   |
| (CO, NO2, SO2) |
+---------------+
        |
        |
+---------------+
| Signal        |
| Conditioning  |
+---------------+
        |
        |
+---------------+
|  ADC          |
+---------------+
        |
        |
+---------------+
| Microprocessor |
```

```
    | /Controller   |
    +---------------+
      |        |
      |        |
+------------+  +--------------+
| Display    |  | Alarm / IoT  |
| (LCD/LED)  |  | Module       |
+------------+  +--------------+
```

Explanation of Each Block

1. Gas Sensors

- Detect pollutants such as **CO, NO_2, SO_2**
- Convert gas concentration into electrical signals

---

2. Signal Conditioning Unit

- Amplifies and filters sensor signals
- Removes noise
- Prepares signal for ADC

---

3. Analog-to-Digital Converter (ADC)

- Converts analog sensor output into digital data
- Enables processing by microprocessor

---

4. Microprocessor / Microcontroller

- Central control unit
- Reads digital data from ADC
- Computes **Air Quality Index**
- Controls display and alarm system

---

5. Display Unit

- Displays AQI value
- Shows pollution level (Good, Moderate, Hazardous)

---

6. Alarm / Communication Module

- Activates alarm if pollution exceeds threshold
- Can transmit data using IoT or GSM

Advantages of AQI System

- Real-time monitoring
- Early warning system
- Low-cost and efficient
- Suitable for smart cities

Applications

- Urban air monitoring
- Industrial pollution control
- Environmental studies

2️⃣ WEATHER FORECASTING SYSTEM

Introduction (WRITE THIS)

Weather forecasting plays a crucial role in **agriculture, aviation, disaster management, and daily life**.
A **microprocessor-based Weather Forecasting System** collects and processes meteorological data to predict weather conditions.

Block Diagram of Weather Forecasting System (MUST DRAW)

```pgsql
```

```
+-------------------+
| Weather Sensors   |
| (Temp, Humidity,  |
| Pressure, Wind)   |
+-------------------+
        |
        |
+-------------------+
| Signal Conditioning |
+-------------------+
        |
        |
+-------------------+
| ADC               |
+-------------------+
        |
        |
+-------------------+
| Microprocessor    |
| /Controller       |
+-------------------+
  |         |
  |         |
+-------------+  +---------------+
| Display Unit |  | Data Storage / |
| (LCD/PC)    |  | Communication  |
+-------------+  +---------------+
```

Explanation of Blocks

1. Weather Sensors

- Temperature sensor
- Humidity sensor
- Pressure sensor
- Wind speed/direction sensor

---

2. Signal Conditioning Circuit

- Amplifies low-level signals
- Filters unwanted noise
- Provides stable input to ADC

---

3. ADC

- Converts analog sensor outputs to digital form

---

4. Microprocessor / Microcontroller

- Reads sensor data
- Processes meteorological parameters
- Performs calculations and forecasting algorithms

---

5. Display Unit

- Displays real-time weather data
- Shows temperature, humidity, pressure

---

6. Storage / Communication Unit

- Stores historical weather data
- Sends data to remote servers or cloud platforms

Advantages of Weather Forecasting System

- Accurate weather prediction
- Real-time monitoring
- Supports disaster prevention
- Useful in agriculture and aviation

Comparison Line (SMART EXAM ADD-ON)

Both AQI and Weather Forecasting systems use **sensors, signal conditioning, ADC, and microprocessor units**, but differ in **type of sensors and application domain**.

SHORT ANSWER (3–4 MARKS)

> A microprocessor-based AQI or Weather Forecasting system uses sensors, ADC, and a processing unit to monitor environmental parameters and display meaningful information.

FINAL EXAM WRITING TIPS (VERY IMPORTANT)

✔️ Draw **neat block diagram**
✔️ Explain **each block in 2–3 lines**
✔️ Use keywords:

- *real-time*
- *monitoring*
- *data processing*
- *environmental parameters*

🎯 YOU ARE NOW FINAL-READY

You have covered:
✔️ I/O Data Transfer
✔️ PPI + CWR
✔️ Handshaking
✔️ PIC 8259
✔️ Double Buffering
✔️ Application Block Diagrams

If you want, next I can:

- 📌 Predict **most likely final paper questions**
- ✍️ Do **answer-writing practice** (I'll check your answers)
- 🧠 Make **1-page revision sheet** for last night

Just tell me 🔥

Excellent catch 👍
This is **exactly examiner-level thinking** — these are the *small-looking but high-impact* theory gaps that cost marks if missed.

I'll now give you **EXAM-READY, BOOK-LANGUAGE answers** for **ALL THREE** things you mentioned, in proper structure.

PART–A: TYPES OF DMA (YOU ARE RIGHT – 3 TYPES)

TYPES OF DIRECT MEMORY ACCESS (DMA)

The DMA controller can perform data transfer using **three different techniques**, depending on how control of the system bus is shared with the CPU.

1️⃣ BURST MODE DMA (Block Transfer / Unconditional DMA)

Definition (BOOK LANGUAGE)

In **Burst Mode DMA**, the DMA controller takes **full control of the system buses** and transfers an entire block of data **without releasing the bus** until the transfer is complete.

👉 This is also called **Unconditional DMA**, because the CPU is completely halted during transfer.

Working

1. DMA controller requests bus control.
2. CPU grants the bus and becomes idle.
3. DMA transfers **entire block of data**.
4. After completion, DMA releases the bus.
5. CPU resumes execution.

Characteristics

- CPU is **suspended**
- Fastest DMA method
- High throughput

Applications

- Disk controllers
- High-speed bulk data transfer

2️⃣ CYCLE STEALING DMA (Conditional DMA)

Definition (VERY IMPORTANT)

In **Cycle Stealing DMA**, the DMA controller **temporarily steals one bus cycle at a time** from the CPU to transfer data.

👉 This is called **Conditional DMA**, because CPU and DMA **share the bus conditionally**.

Working

1. DMA controller requests bus for **one cycle**.
2. CPU pauses for that cycle.
3. One byte/word is transferred.
4. CPU resumes execution.
5. Process repeats.

Characteristics

- CPU execution is **slowed but not stopped**
- Medium speed
- Balanced performance

Applications

- Audio systems
- Communication devices

3️⃣ TRANSPARENT DMA (Hidden DMA)

Definition (BOOK LANGUAGE)

In **Transparent DMA**, the DMA controller transfers data **only when the CPU is not using the system buses**, making the DMA operation invisible to the CPU.

Characteristics

- CPU operation **not affected**
- Slowest DMA method
- Used in low-priority transfers

Applications

- Background data transfer
- Low-speed peripherals

DMA TYPES COMPARISON (EXAM GOLD)

DMA Type	CPU Halt	Speed	CPU Impact
Burst (Unconditional)	Yes	High	CPU stopped
Cycle Stealing (Conditional)	Partial	Medium	CPU slowed
Transparent	No	Low	CPU unaffected

PART–B: PROGRAM DATA TRANSFER

CONDITIONAL vs UNCONDITIONAL

UNCONDITIONAL DATA TRANSFER

Definition

Unconditional data transfer instructions **always execute**, regardless of the condition flags.

Examples

- JMP
- STA
- LDA
- MOV
- MVI

Key Point (WRITE THIS LINE)

> Unconditional instructions do not depend on the status of flags such as Zero, Carry, or Sign.

CONDITIONAL DATA TRANSFER

Definition (VERY IMPORTANT)

Conditional data transfer instructions execute **only when a specific condition flag satisfies the given condition**.

Common Conditions

Condition	Flag
Zero	Z
Not Zero	NZ
Carry	CY
No Carry	NC

Examples

- JZ, JNZ
- JC, JNC
- CZ, CNZ

Key Exam Line

> Conditional instructions enable decision making and program control based on computation results.

PART–C: 3-BYTE INSTRUCTIONS + T-STATES (VERY IMPORTANT)

Teacher **LOVES this table**.

WHAT IS A 3-BYTE INSTRUCTION?

Definition

A **3-byte instruction** consists of:

- **1 byte opcode**
- **2 bytes operand or address**

COMMON 3-BYTE INSTRUCTIONS (8085)

1️⃣ STA (Store Accumulator Direct)

Property	Value
Bytes	3
T-states	**13**
Function	Stores A to memory

2️⃣ LDA (Load Accumulator Direct)

Bytes	3
T-states	**13**
Function	Loads A from memory

3️⃣ JNC (Jump if No Carry)

Bytes	3
T-states	**10 if jump, 7 if not**
Condition	CY = 0

4️⃣ JNZ (Jump if Not Zero)

Bytes	3
T-states	**10 if jump, 7 if not**
Condition	Z = 0

5️⃣ JMP (Unconditional Jump)

| Bytes | 3 |
| T-states | **10** |

6️⃣ LXI H (Load HL Immediate)

Bytes	3
T-states	**10**
Function	Loads 16-bit data into HL

7️⃣ ANI (AND Immediate)

Bytes	2 ❗ (IMPORTANT TRICK)
T-states	**7**
Function	Logical AND with accumulator

⚠️ Many students mistakenly write ANI as 3-byte — **it is 2-byte**.

NOT 3-BYTE (BUT YOU MENTIONED THEM – GOOD TO CLARIFY)

Instruction	Bytes	T-states
CMP r	1	4
DCR r	1	4
MVI r	2	7
BACK (label)	Depends on instruction	

👉 Writing this clarification in exam shows **strong conceptual grip**.

SUMMARY TABLE (VERY EXAM FRIENDLY)

Instruction	Bytes	T-states
STA	3	13
LDA	3	13
JMP	3	10
JNC	3	10 / 7
JNZ	3	10 / 7

Instruction	Bytes	T-states
LXI H	3	10
ANI	2	7

FINAL EXAM TIP (VERY IMPORTANT)

If teacher asks:

> *"Explain with example and timing"*

You should:
✔️ Define instruction
✔️ Write bytes
✔️ Write T-states
✔️ Mention condition (if any)

🔥 You are now FULLY COVERED for MBSD finals

If you want next, I can:

- 📌 Predict **exact final questions**
- 📝 Take **mock written answers** and correct them
- 📄 Make **last-night revision sheet (2 pages)**

Just tell me 💪